

The Social Costs of Tobacco Smoking

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Summary

This article is concerned with establishing a methodology for the measurement of the social cost of tobacco consumption, and with using that methodology to estimate the relative quantitative importance of each of the identified components of social cost. It begins by defining social cost and explaining its relevance to policy making. The next section derives a matrix of items that need to be calculated if the social cost is to be correctly measured. The third section surveys the existing empirical literature on the social costs of tobacco consumption in the light of the methodology derived in this paper. In general it concludes that most measures of 'social cost' are incorrect. However, some interesting numbers do emerge from the studies and these are highlighted. Finally, the last section discusses the policy relevance of actual and potential measures of social cost and concludes with some general observations on the direction of future research in this field.

Introduction

The social cost of smoking tobacco is a subject on which there is a great deal that has been written and said but which is littered with conceptual confusion, inappropriate measurement and invalid inference. The purposes of this article are: to show where the errors in the existing methodologies lie; to try and clear some of the confusion that is present; and to indicate how one might go about measuring this cost correctly, using the most recent advances on the economic and statistical fronts. The material in the article is organized as follows. In this section the basic notion of social cost is introduced and its relevance to policy making is explained. In the next section the items that constitute the social cost of tobacco consumption are examined individually and the issues that arise in measuring them are reviewed. This leads to a typology of social cost that is presented in a matrix. In the third section the existing empirical literature on the social cost of tobacco is reviewed with reference to the methodology established in the previous section. In general the conclusions to be drawn are fairly negative, with most of the cost estimates being invalid. However,

there are some interesting numbers that emerge from this exercise and these are highlighted. The final section discusses the policy relevance of actual and potential measures of social cost. The paper is concluded with some general observations on the direction and value of future research in this area.

The Concept of Social Cost

Every human activity that can be described as one involving the consumption or production of goods and services generates certain costs. These arise because the activities use up scarce resources which would have some value in other uses. To the extent that the costs are knowingly and freely borne by the consumer or producer himself, they are referred to as PRIVATE COSTS but to the extent that they are not so borne but fall on the rest of society, they are referred to as SOCIAL COSTS. Hence the total cost of any activity is the sum of the private and social costs.

The simple distinction, however, is complicated by a number of factors. As far as tobacco consump-

tion is concerned, the most important of these is the extent to which the consumer is *aware* of the costs that he bears. If his actions are determined by a perceived cost that is in fact less than his actual cost, then the difference between the two can be viewed as a social cost. The reason for making this interpretation is that the individual himself has not adjusted his behaviour to reflect these higher costs and they are therefore, in some sense, unaccounted for. It is this unaccountable nature of some of the costs that are incurred in the course of economic activity that one is trying to capture in the definition of social cost. The implications of this definition will become clearer when specific items of social cost arising from tobacco consumption are looked at later in this article.

The second key, and related, issue that arises in connection with tobacco consumption is that of addiction. There are different ways of characterizing addiction and these are discussed below. However, they all raise the difficult question of what it means to *knowingly* incur certain costs. With a 'rationally' addicted individual—i.e. one who knows how his desires and tastes will change in the future as a result of his smoking now—one can speak of costs having been knowingly and willingly incurred. But such cases are rare (or even non-existent) and so the problem of how to measure social costs in the presence of addiction will remain.

The Implication of Social Costs for Social Policy

When the social costs of tobacco consumption have been measured and found to be £X or Y million per annum, the question that arises is what implications do they have for regulatory policy. The fundamental point to be made here is that such policy must be determined as a result of the balance of the social and private costs of tobacco consumption against the benefits of smoking to the consumers themselves. Each specific policy would have to be evaluated within a cost-benefit framework and the decision to implement it made according to whether it generated a positive *net benefit*. This would mean that the reduction in social and private costs was less than the loss of benefit. The existence of a large social cost item in the aggregate does not guarantee that this will always be the case.

The Social Costs of Smoking—A Conceptual Framework

The only previous attempt to classify social costs

according to a proper economic framework is that of Atkinson & Meade (1974). The Atkinson-Meade analysis is essentially correct, but in what follows some amendments and extensions to it have been suggested.

The types of cost that are widely considered to be constituent parts of the social cost of smoking are listed below, with each one given an acronym.

- (i) Costs of medical treatment for smoking related diseases. [MEDTR]
- (ii) Loss of earnings during periods of sickness related to smoking. [SNERNLOSS]
- (iii) Social security benefits paid out to smokers during periods of smoking related illness. [SNBEN]
- (iv) Cost to the individual smoker of premature death. [VOL]
- (v) Cost to the individual of pain and suffering due to smoking related disease. [VOSUFF]
- (vi) Transfer payments such as pensions and taxes, paid or forgone due to premature death. [BENTAX]
- (vii) Value of the resources used up in the production of tobacco products (and in tobacco growing if the analysis is extended to other countries). [TOBPROD]
- (viii) Taxation of tobacco products. [TOBTAX]
- (ix) Losses of well being, pain and suffering to relatives and close friends. Such losses arise from anxiety about the risks a smoker is running, the observation of, and sympathy with sufferer due to smoking related illness, and grief and suffering at the premature death of a smoker. [RELWEL]
- (x) Externalities to non-smokers. These will include any costs of 'passive' smoking, any annoyance caused (e.g. coughs, eye irritation) to non-smokers, fire risks, cleaning of garments, litter etc. [EXT]

For each of these items, some proportion constitutes a component of social cost, but they *cannot* simply be added to produce a total. Some are relevant in some circumstances and some in others. They are discussed in detail below.

Medical Treatment Costs [MEDTR]

It is now generally accepted that the medical costs of tobacco consumption should be measured over the lifetime of an individual. Since this means that future savings which would result from the premature death of a smoker have to be credited to the

overall medical costs, the total value of MEDTR could be negative, zero or positive. The second point to note is that, whatever the value, it is not relevant to the social cost calculation if it is incurred through a private insurance or direct payment scheme. This produces the immediate observation that virtually all social cost studies are in error in simply calculating a proportion of health care costs as 'due to' smoking related diseases and failing to account for future savings in costs. The only exceptions are the studies of Atkinson & Townsend (1977) and Leu & Schaub (1983, 1984). It is notable that both these studies come up with very small savings from reduced medical treatment costs.

Sickness Earnings Loss [SNERNLOSS]

This relates to the losses in earnings due to smoking related sickness. If the smoker is informed about the risks of smoking, such losses will be anticipated by him and the foregone wages will be 'internalized' as part of the private cost of smoking. In such circumstances SNERNLOSS is not relevant to social cost calculations. However, if the smoker is inadequately informed of the true risk, then part of SNERNLOSS will constitute a social cost. The correct measurement of such a social cost requires the demand curves for tobacco to be identified with and without the correct information. However, this can be approximated by multiplying the difference between the true and the perceived probability of income loss by the loss of earnings itself. It should also be noted that if SNERNLOSS is relevant, any transfer payments received because of the sickness—e.g. sickness benefit [SNBEN]—must not be added to SNERNLOSS. Such benefits are part or whole compensation for the loss of earnings and adding the two would be double counting.

It is also important to distinguish between SNERNLOSS from any foregone output due to premature death. Such estimates are common in social cost studies, but only in a few cases does there appear to be any discussion of what such forgone earnings might be measuring. This issue is returned to when the question of the value of life is discussed.

Sickness Benefit [SNBEN]

If the smoker is aware of the health risks of smoking, he will have internalized any sickness related foregone earnings. But if in these circumstances he still receives sickness benefit that benefit, albeit a transfer payment, is a cost to society. This is

because his private cost will have been lowered by the amount of the benefit and he will be operating on the basis of that lower cost. Thus SNBEN is relevant and positive in the awareness case. In the unawareness, however, SNBEN is not relevant because of the double counting argument given in SNERNLOSS.

Value of Life [VOL]

Placing a money value on life may seem irrational or morally illicit. But no society operates as if life had an infinite value. Instead one is quite used to the idea of placing a money value on avoiding a small risk of death—as with the cost of seat belts, vaccinations, crash barriers and so on. Such expenditures imply a value for saving a life. If smokers are aware of the health risk they run by smoking then they will take account of the risk of death in their decisions on whether to smoke or not. Thus, for the rational and informed smoker, the value that he places on his own life is not relevant to the calculation of social cost.

In the limited information case, smokers will understate the risk to their own health and what they lose from the unrealized risk will be the value they place on their own lives, i.e. VOL. However, it should be noted that none of the studies surveyed here makes use of the literature on VOL. Instead authors typically confuse several concepts and relate them to foregone production due to mortality. The concepts that are confused are the smoker's valuation of his own life—what is called VOL here—and some measure of cost to society in terms of 'lost' gross national product (GNP) because of premature death. As noted above, the former is relevant to the social cost of smoking to the extent that smokers can be judged to be uninformed about the health risks. The latter, however, is not a social cost, at least not wholly. This is because society as a whole does not 'lose' any GNP simply because smokers die prematurely in comparison with non-smokers. It only 'loses' what the smoker would have paid to the rest of society in the form of taxes and 'gains' what it would have paid to him through transfer payments. Both calculations should be carried out on a lifetime basis and when this is done the net discounted figure may turn out to be a cost or a benefit to the rest of society.

One possible source of confusion in this area arises because authors frequently use foregone production to measure VOL. In doing so they have a good historical precedence from other areas of work

on the value of life. Until recently this was measured by the value of production that was expected to ensue had the person at risk lived to a 'normal' age. This approach is now regarded as unsatisfactory because the calculated value has no basis in the *individual* valuation of life. The latter is based on the concept of willingness to pay to reduce the risk of death by a specified amount and, although there are some theoretical problems with it this method too, it is broadly accepted as the correct approach.

Empirically this valuation has been carried out by looking at differences in wages among workers and identifying that part of the wage that is attributable to the riskiness of the occupation. The technique has shown considerable success in obtaining risk related wage differentials, although the figures across studies do not always agree. Details may be found in Violette & Chestnut (1983). Once the risk premium has been identified it can be used to calculate the cost to the individual of an increased risk of death in other situations such as those arising from smoking. This transposition of risk premia from one situation to another, however, is not always acceptable and in doing so one should be mindful of the fact that the risk level at which the premium is calculated is important. Ideally one should choose studies carried out at a similar risk level as that incurred in smoking.

Although it would be interesting to see a study of the costs of smoking that made use of such data, unfortunately none exists. Hence it remains an unanswered question whether the numbers that would emerge from such an exercise would be smaller or larger than those that are obtained from a productivity approach.

The Smoker's Own Suffering [VOSUFF]

If smokers are aware of the risks of smoking they will allow for any suffering due to smoking related ill health in their private cost-benefit decision. To the extent they are unaware, their suffering will constitute an unanticipated cost. Care has to be taken, however, not to double count this value of suffering. VOSUFF has to be net of any privately borne health costs. It has also to be net of any value of life (VOL) calculation. For an up-to-date discussion of alternative models of the measurement of such morbidity costs see Berger *et al.* (1985).

None of the social cost of smoking studies surveyed contains any empirical measure of VOSUFF. Leu & Schaub (1984) provide an indica-

tor of 'years of life with impaired life quality' for Switzerland but, as noted above, this may or may not include genuine social cost.

Lifetime Benefits and Taxation [BENTAX]

BENTAX measures the difference between the smoker's contributions to the national exchequer through taxation and the amounts he withdraws from that exchequer through various forms of social security payments. These amounts are calculated over the *lifetime of the non-smoker* in accordance with the principles outlined previously. That is, the taxes and benefits that would have been paid if the smoker had lived to a normal age have to be estimated and a discounted present value of the two streams of financial flows calculated. If the net present value of the benefits exceeds that of the costs, there is a net social cost from smoking and vice-versa.

This item of social cost/benefit is relevant in all cases, i.e. whether the smoker is aware or unaware or whether he is addicted or not addicted.

In practice the empirical studies that are undertaken fail to follow the methodology outlined above for estimating BENTAT. The main reason for this is their failure to look at the lifetime approach to calculating social cost. An exception is the study by Leu & Schaub (1984), which separates out lifetime costs for health insurance, social security and disability insurance, employer based pension schemes and sickness benefits. They find that the estimated costs due to BENTAX are about three times the estimated incidence of medical costs (MEDTR in this paper).

Tobacco Taxation [TOBTAX]

The taxation of tobacco products is widely referred to in debates about the social costs and benefits of smoking, primarily because pro-smokers argue that society benefits more from tobacco taxation than any social costs imposed by non-smokers. In fact the role played by TOBTAX is not at all straightforward. In large part this is because of differences of opinion as to the *purpose* of such taxes. At one extreme is the view that the tax is explicitly designed to deter smokers from smoking and/or to recompense society for the social costs of smoking. Such a tax would be referred to in economic terms as the Pigovian tax, after the economist, A. C. Pigou, who elaborated the idea of taxing products according to the social costs they impose.

If the 'Pigovian' approach is adopted, as Atkinson & Meade (1974) have done, then TOBTAX has to be *deducted* from any measure of social cost because it amounts to compensation being paid by smokers to the rest of society for the damage they do. This means that the measured social cost is a *net cost*, which takes account of the benefits generated by the compensatory payments. However, indeterminacy enters the picture if TOBTAX is not seen to be a pure Pigovian tax. There is in fact very little evidence that excise duties on tobacco began their existence as such taxes. Indeed, the presumption must be that excise duties on tobacco owe far more to the fact that the cigarette smoking is price inelastic (increasing price does not reduce demand in equal proportion). However, there is some evidence to suggest that the increased taxes on tobacco in the 1970's and 1980's have at least partly been justified on Pigovian grounds. If so, a *proportion* of TOBTAX should be deducted from gross social cost to obtain a net cost figure. This is the view taken in this paper.

The Welfare of Relatives [RELWEL]

Addicted, uninformed or otherwise, if a smoker does suffer some disease because he smokes, there will be suffering caused to relatives and near friends. The same applies to premature death. This grief and suffering is a proper social cost.

Atkinson & Meade (1974) record this as a cost only in the case of the unaware and unaddicted smoker. This is not the view taken here since it is difficult to argue that suffering of relatives is in any way different simply because the individual is unaware or addicted.

Other Externality [EXT]

If there are costs to non-smokers in the form of passive smoking, annoyance, fire risks, and so on, all such costs can be regarded as 'proper' social costs and they should be added to the other costs. The empirical literature makes occasional reference to the costs of fires started through smoking. No attempts are made to value annoyance to non-smokers. Epidemiological studies have tried to estimate passive smoking deaths, but no valuation has been made in money terms.

Costs of Producing Tobacco Products [TOBPROD]

Producing tobacco products uses up resources

which have an obvious economic value. However, the costs of these resources are not social costs in the sense defined here because they are incurred for the purpose of providing a good which yields private benefits. That is, making cigarettes imposes no more social cost in this sense that does the manufacture of strawberry jam. However, as will be shown below, there is one case in which TOBPROD is relevant to the calculation of social cost.

A Matrix of Social Cost

The discussion on the conceptual basis for the calculation of social cost can now be summarized as shown in Fig. 1. This is most conveniently done using a matrix, suggested by Atkinson & Meade (1974), but amended and modified in the light of the above analysis. In particular: VOSUFF has been added to the items of social cost; the argument that all TOBTAX should be deducted from gross social cost has been modified; RELWEL has been added in all cases; and VOL has been introduced, which seems to have been subsumed under 'costs of premature death' in the Atkinson analysis. Most of the matrix is self-explanatory in terms of the preceding discussion, but some observations are needed.

(a) Fig. 1 distinguishes four cases, following the addicted/unaddicted and informed/uninformed categories. Along with Atkinson & Meade, the view that the aware case is the more empirically relevant is accepted as broadly correct. In that case it is boxes 3 and 4 that are relevant.

(b) Box 3 presents problems of interpretation since it is not at all clear what awareness of risks combined with addiction actually implies. Atkinson & Meade (1974) similarly leave this box empty. However, it should be noted that the notion of an 'aware addict' is not meaningless: smokers do sometimes report that they know the publicized risks of smoking but are 'unable' to give up.

(c) the α in front of the TOBTAX box indicates that some proportion of TOBTAX is being deducted, where α relates to the proportion of the tax that is judged to be due to Pigovian motives.

(d) Boxes 1 and 2 include earnings loss (SNERNLOSS) and value of life (VOL). These arise because of the unawareness of the consumer of the health risks. It should be noted that these items have to be multiplied by the difference between the true probability and the perceived probability of incurring them in order to obtain an appropriate

	ADDICTED	NOT ADDICTED
UNAWARE	MEDTR+VOL+ SNERNLOSS+ BENTAX+ RELWEL+VOSUF+ TOBPROD+EXT (1)	MEDTR+VOL+ SNERNLOSS+ BENTAX- α TOBTAX+ RELWEL+VOSUF+ EXT (2)
AWARE	?? (3)	MEDTR+SNBEN+ BENTAX- α TOBTAX+ RELWEL+EXT (4)

Figure 1. A matrix of social cost.

measure of cost to be added to the other measures in those boxes.

(e) Box 1 requires some further explanation since it is quite different to the others in *adding* TOBPROD as a social cost. It relates to the polar case where the smoker is both addicted and unaware. The interpretation of this box is that suggested and unaware. The interpretation of this box is that suggested by Atkinson (1974) in which the addict derives no pleasure at all from smoking. Such a smoker has wholly distorted choices because he is purchasing something that gives him no benefit. His total expenditure on tobacco is thus an 'overvaluation' and the smoker could benefit by an amount equal to that expenditure if he spent it on something else. His expenditure on tobacco is equal to the cost of producing it (including tobacco companies' profits) and the excise tax. Thus TOBPROD+TOBTAX is the gross cost of addiction. But TOBTAX is a transfer payment (i.e. it accrues as a benefit to others). So the true cost of addiction is TOBPROD only. To this must be added the other social costs. However, whether addiction can be characterized in this way is open to debate. As was pointed out at the beginning of this paper, there is a view of a 'rational' addict. Such a person takes account of how his tastes will change as a result of consuming a particular product and, for him, there are no direct social costs to consider. When habit formation is present in conjunction with unawareness, however, the problem has to be looked at from an informational perspective and, in that case the use of TOBPROD would imply that if such a consumer could be made aware of the risks he would give up completely. Since this unlikely to be true,

TOBPROD probably overestimates the social costs of this aspect of tobacco consumption.

This matrix of social cost provides a 'measure' against which to judge the various empirical studies. These are looked at in the next section.

The Social Costs of Smoking: the empirical literature

This section comments on the individual empirical studies that have tried to quantify the cost items discussed in the last section. The comments presented here are in general terms and try to indicate the main issues that arise in each area of quantification.

Medical Costs

Table 1 shows money values of health care costs from various studies and which costs are argued by the individual authors to be attributable to smoking. The estimates are standardized in terms of 1980 U.S. dollars, using official exchange rates. The basic methodology for estimating medical costs is straight forward. For a given population, the mortality ratio is estimated as deaths from a given disease for smokers relative to non-smokers. Combined with the numbers in the population who smoke relative to the total population of smokers and non-smokers combined, it is then possible to calculate an 'attribution' factor which measures the proportion of deaths from that disease and which can be regarded as arising from smoking. Given that the number of deaths due to a smoking related disease are now estimated, this number is multiplied by some treatment cost per disease. Substantial vari-

Table 1. Comparative Medical Care Costs

Author of study	Region covered	(1980 U.S. \$)		
		Date of estimate	Total cost (millions)	Cost per smoker
Shillington [1977]	Canada	1971	98-143	16-24
Stoddart ^{1,2} [1986]	Ontario	1978	21-42	8-16
		1978	157	59
		1982	225	84
		1979	1080	108
Collishaw and Myers [1984]	Canada	1979	1080	108
Thompson and Forbes [1982]	Canada	1980	2008	199
Hjalte [1984]	Sweden	1980	119	48
Leu and Schaub ³ [1984]	Switzerland	1976	25-51	13-27
Yach [1982]	South Africa	1981	23	3
UK DHSS ² [1972]	U.K.	1970	299	14
UK DHSS ^{1,2} [1985]	England	1985	420-756	25-45
Cohen ² [1984]	England & Wales	1981	330	18
Kristein [1977]	U.S.A.	1975	8070	367
Luce and Schweitzer [1978]	U.S.A.	1976	11843	197
U.S. Office of Technology Assessment [1985]	U.S.A.	1985	8350-24340	167-487

Notes: ¹ See text for explanation of different figures.

² Public expenditure only.

³ Uses lifetime costs and savings approach.

Source: computed and adjusted by the authors from the original data.

ability in the resulting estimates can be expected according to:

- (1) which items of cost have been included in the estimates
- (2) what diseases are thought to be smoking related
- (3) whether attribution factors are estimated for populations as a whole or sub groups within that population
- (4) differing attribution factors arising from differing mortality and morbidity ratio estimates
- (5) different estimates for the cost of treating the disease
- (6) whether or not the 'life cycle' approach has been adopted

Estimates of the cost of health care attributable to smoking (in Table 1) vary from a low of \$3 per smoker per year (Yach (1982)) to \$487 per smoker per year (Office of Technology Assessment, 1985). Particular points to note are:

(a) The estimated cost for South Africa is the lowest because it only includes hospital costs whereas the others include physician costs. Even this figure is an under-estimate of what it seeks to measure because it is based on the average cost of a day in hospital and average length of stay. Both

these are under-estimates of the corresponding items in the case of smoking induced illness.

(b) The Stoddart *et al.* (1986) study uses the basic attribution factors collected by Shillington (1977) on 1978 data. Their most comparable estimates to those of Shillington are the first range of \$8-16 per smoker. This is lower than the range obtained by Shillington, which is surprising and cannot be explained from the published information available. The higher figures given by Stoddart *et al.*, for 1979 and 1982 are based on a wider set of associated costs and related diseases than the earlier Canadian study.

(c) Collishaw & Myers (1984) produce a cost for Canada in 1978-1979 that is \$108 per smoker per year. Their figure is relatively higher than the others because they use different, and more accurate 'attribution factors' than the others. They also have more detailed data concerning the relative use of physician services by smokers and non-smokers and on the 'disease set' that is associated with smoking. As a result of all these factors their *overall* cost of smoking is the highest of the Canadian studies.

(d) While Thompson & Forbes (1982) have a lower overall cost of smoking their medical cost is higher than that of Collishaw & Myers: \$199 per smoker, compared to \$108 per smoker. This is partly due to their inclusion of other facets of health care but this cannot explain the whole difference.

(e) As indicated earlier Leu & Schaub (1984)

estimate the costs of smoking in a manner that is closest to the methodology presented in this paper. The first difference with the earlier studies is that they take a lifetime view of the costs of smokers versus non-smokers. Although smokers have higher annual medical costs, they have lower life expectancy and thus generate savings in medical costs which partly offset the costs of treating smoking related diseases.

Leu & Schaub also recognize that smokers have other characteristics that non-smokers do not have and which may account for some of their diseases. For example, smokers tend to drink more and therefore have a higher incidence of cirrhosis of the liver. The study attempts to take these factors into account and concludes that 65% of smokers' excess mortality can be attributed to smoking *per se*. By not recognizing this factor, other studies over-estimate the costs associated with smoking.

(f) Hjalte (1984) uses the Leu & Schaub methodology for Sweden and obtains a figure of \$48 per smoker, against \$13-27 by Leu & Schaub. It is not possible to account for the difference between the two studies.

(g) For the United Kingdom, there are three sets of estimates, ranging from \$14 to \$45 per smoker. Each estimate uses more recent data on attribution rates and therefore reflects more up-to-date assessments of the relationship between smoking and illness. The last two figures, however, include only hospitalization and physician costs whereas the first one includes pharmaceutical costs as well.

(g) All the figures for the United States are high in comparison with studies for other countries. They range from \$167 to \$487 per smoker per year. Both these extreme values are from the Office of Technology Assessment (OTA), which probably represents the best of the U.S. studies. The wide range of cost estimates in this study arises from the use of substantial variability in the estimates of attribution factors. As these arise from uncertainty in the epidemiological rather than the economic data there is little that can be said about them in the context of this paper, except to point out that if the uncertainty is believed to be so high in the U.S. sample, it cannot be much less in the data from the other countries. Nevertheless, these other country studies appear generally to come up with much smaller ranges of estimates.

Conclusions on Medical Costs

It is difficult to explain the wide variation in the

estimates of MEDTR. Much depends on the precise source of data and methods of estimation. The Leu & Schaub and Hjalte estimates are the only ones that use a life cycle approach, and these produce relatively low figures. However, the DHSS figures overlap the Swiss ones and the highest U.K. figure is actually one tenth of the highest U.S.A. figure. Differences in the cost of treatment will explain part of the differential, as will variation in the analysis relating to the disease set and attribution factors.

Value of Life/Lost Production

A major item in social cost studies of smoking is the loss of welfare due to illness and premature death. As noted earlier, most of the empirical literature seeks to measure this cost by using a 'foregone output' concept. However, this needs more careful interpretation than it has received in the past. In the second section it was noted that:

- (i) earnings losses (SNERNLOSS) are relevant to the social cost of smoking if the consumer is unaware of the risks. If he aware, this cost is irrelevant;
- (ii) past literature assumed a close relationship between the value of life (VOL) and forgone output. If that literature were correct, then foregone output due to early mortality would measure VOL, and VOL was seen to be relevant in the case where the smoker could be deemed to be unaware of the risks of smoking.

In both these cases then, lost output is not a measure of social cost if the smoker is 'reasonably informed' about the risks of smoking. In the second case, the proper measure is VOL and the state of the art in the theory of social cost tells us that VOL is not systematically related to foregone production due to premature death.

In the second section, the idea that changes in GNP or lost production can be a measure of the costs of smoking was discussed and it was shown that, in general, they are not. An interesting study by Richter & Gori (1980) simulates the effects of a successful anti-smoking program in the U.S.A. They show that such a program would lead to reduced GNP (some \$16 billion by the year 2000 in 1972 prices), and increased unemployment. Hence it clearly demonstrates that the implication that reducing smoking will increase GNP is incorrect.

Although the conceptual basis for the foregone output approach is unsatisfactory, its numbers are

Table 2. Value of 'Lost' Production due to Mortality and Morbidity

Author of study	Country/ date	(1980 U.S. \$)			
		Mortality		Morbidity	
		Total cost (millions)	cost per smoker	Total cost (millions)	cost per smoker
Shillington [1977]	Canada 1971	465	76	348-436	57-72
Collishaw and Myers [1984]	Canada 1979	3094	310	576	57
Yach [1982]	South Africa 1982	264	30	8	0.85
Hjalte. [1984]	Sweden 1980	457	191		
Leu and Schaub [1984]	Switzerland 1976	216-265	114-140	110-190 ¹	58-101 ¹
Kristein [1977]	U.S.A. 1975	18000	300	4650	78
Luce and Schweitzer [1978]	U.S.A. 1976	18645	311	8916	149
O.T.A. [1985]	U.S.A. 1985	20520-46360 ²	410-927 ²		

Notes: ¹ Includes rehabilitation and disability pensions.

² Include both mortality and morbidity losses.

of interest because they have an influence on policy towards smoking. Hence they have been included in this paper. Table 2 summarizes the empirical measures, which are standardized in 1980 prices, and which attempt to distinguish between morbidity costs (SNERNLOSS) and mortality costs (for which there is no counterpart in the methodology presented here). The estimates range for losses in mortality range from \$30 per smoker (Yach) to \$311 per smoker (Luce & Schweitzer). Morbidity costs range from \$1 to \$149 while combined mortality and morbidity costs range from \$31 to \$927 (the OTA study sums the two). These differences will obviously partly reflect income differentials as figures are based on average earnings within the country in question.

Externality

Passive Smoking. It is increasingly argued that inhalation of smoke by non-smokers who are habitually in the presence of smokers causes illness and premature death in the non-smoking population (Correa *et al.* (1983), Hirayma (1981), Trichopoulos *et al.* (1981, 1983), among others). Repace & Lowrey (1985) suggest that passive smoking may be responsible for one third of lung cancer deaths in non-smokers in the U.S.A. The statistical basis of this estimate is debated in respect of the non-random nature of mates—smokers marry smokers

(Burch, 1981)—limited sample size and difficulty of adequate statistical control.

If passive smoking occurs and if it is related to exacerbated illness or to death in non-smokers, then it constitutes a very clear example of an external cost of smoking. The principles of measuring such an external cost are clear enough but, unfortunately, no attempt has been made to do this so far.

Fires. Table 3 shows various estimates of the costs of fire damage which are alleged in the studies to be attributable to smoking. The estimates are for the United States and Canada only. They vary from \$8 million to \$510 million per year.

Strictly speaking what is required in such studies is a value of life figure for fire victims and a property value for the fire damage done. For Canada, Shillington's (1977) estimate of \$8 million includes only the foregone income of the fire victims, as a proxy for value of life lost. Collishaw and Myers (1984) estimate of \$778 million excludes the former, but includes the costs of fire fighting and the loss of forest and property.

In the United States the estimates of the cost of fire damage due to smoking range from \$248 million to \$510 million per year. McGuire (1983) bases his estimates solely on property damage, and his figure of \$248 million is the lowest of the U.S. estimates. The highest estimate for fire costs is produced by Kristein (1983). Unfortunately it is unclear from

Table 3. Costs of Fire Damage

Author of study	Region covered	Date	(1980 U.S. \$)		
			Fires attributable to smoking (%)	Estimate of cost of fire damage	Fire damage cost per smoker ³
Shillington [1977]	Canada	1971	9.2	8	
Collishaw and Myers [1984]	Canada	1979	16.0 ¹	78	9.8
Luce and Schweitzer [1978]	U.S.A.	1975		253	5.3
McGuire [1983]	U.S.A.	1981	45.0 ²	248	6.3
Kristein [1983]	U.S.A.	1980	7.3-11.0	338-510	11.7

Notes: ¹ Forest fires only.

² Fatalities only.

³ Smoking population: U.S. figures based on 50 m. in 1985, 60 m. in 1975; figures at dates between are extrapolated.

Canadian figures based on actual data.

his article exactly what items of cost have been included in his estimates.

It is worth noting that when the above estimates are calculated on a per smoker basis, the figures are much closer, ranging from \$5.3 to \$11.7 per smoker per year.

An Alleged Social Cost: Research. Some writers treat costs of research into the effects and treatment of smoking as social costs. This is not conceptually correct since such costs represent part of the cost of policy options to correct any divergence between private and social cost in a cost-benefit appraisal of anti-smoking policy. Freckler & Ashley (1983) compute such costs at \$2 million per year for Ontario alone, or just under \$1 per smoker per year. Numerous other data are available on the costs of anti-smoking campaigns, but, as noted, these are not social costs.

Workplace Costs. Kristein (1983) suggests that the costs to one company in the United States amounted to between \$306 and \$547 per smoker per year (1980 U.S. dollars). This total is made up of excess insurance costs, fire, workers' compensation, absenteeism, productivity loss of smokers and non-smokers and costs of occupational health. Loss of productivity arises from time lost to smoking rituals, extra clean up costs, damage to equipment, furniture and fixtures, extra inefficiency and errors due to high level of carbon monoxide in smokers' blood which causes lower attentiveness, and cognitive ability and exercise capability reductions.

Weiss (1981a, 1981b) estimates costs per smoker to industry to be significantly higher than Kristein: \$4196 per smoker per annum. This estimate is

calculated by taking into consideration higher medical costs (incurred by the company); insurance considerations; time lost through smoking rituals; burns and damage to furniture; maintenance and cleaning costs; and adverse effects on the morale of non-smokers. Weiss concludes by calculating that if a company hired only non-smokers, it could reap the benefits by having 10% fewer employees doing the same amount of work.

Studies have been carried out on companies with smoking cessation programmes. Repace (1981) cites two examples in his article. Dow Medical Department found that smokers used 80% more sick leave than non-smokers, which cost the company \$494,000 per year. Marle Normal Cosmetics forbade its employees to smoke and saved \$26,000 in reduced sick leave and housekeeping, and in increased productivity.

Which of these costs are *social* costs is difficult to assess. On one interpretation, a perfectly functioning labour market would internalize all such costs through wage-bargaining procedures. That is wages would be adjusted downwards compared to what they would otherwise be to reflect the costs imposed by smokers. Although wages in general may well reflect such costs, there is no evidence that *differential wages* are paid to smokers versus non-smokers. Hence part of the costs of smoking are borne by the smokers but part are borne by the rest of the workforce. It is this latter that constitutes a social cost, but not the entire cost, as taken by Kristein & Weiss. However, their figures can be seen as an upper bound to this item of social cost.

Relatives Welfare. As argued in the second section, this is a legitimate component of social cost. The

only study to compute a relationship between relatives' welfare and the value of life of the person at risk is that of Needleman (1976). The research suggested that, for the U.K. in 1974, the ratio of relatives' valuation to the value placed by the person at risk was 0.45. To put it another way, if the value of life (VOL) is known, it can be multiplied by 1.45 to obtain a composite valuation of the person at risk and his relatives. This ratio varies with a 'coefficient of concern' which in turn is based on analysis of actions taken to reduce risk of death by increasing one's own risk (e.g. donating kidneys). Finally, it should be noted that even though VOL is not relevant to social cost in the 'fully aware' case, the relatives welfare (RELWEL) is. It seems appropriate, then, to use Needleman's ratio to obtain RELWEL in the case of 'aware' smokers by taking 45% of what VOL would have been if the smoker was unaware. No such methodology has been attempted in the literature.

Conclusions on the Empirical Literature

The overriding conclusion on the empirical literature must be that the various estimates produced are generally without foundation in any adequate theory of social cost. Authors frequently secure estimates without asking what it is they are measuring, and certainly without reference to a stated methodology. As such, the literature is, by and large, of very limited value in contributing to the debate about smoking. This is unfortunate because some of the contributions do point the way to a proper cost methodology, notably the earlier work of Atkinson (1974). In spite of these comments, the survey is useful in illustrating the kinds of procedures that have been used, the pitfalls of failing to distinguish private and social cost, and the items that are regularly omitted, e.g. costs to relatives, the value of life. It must not be assumed therefore that a 'proper' study would result in estimates below those given in the literature. *A priori*, there is no way of knowing what such a study would show. But it is strongly recommended that such a study be carried out.

Conclusions

This paper has identified the main items of social cost arising from smoking tobacco, discussed the conceptual issues involved in their measurement and recommended certain procedures, and examined the empirical evidence in the light of these recommendations.

In general it was concluded that there are several misconceptions in the way that the costs of smoking are conceived and several errors in the way they are empirically measured. How far these misconceptions and errors matter depends very much on what use is made of the studies for purposes of social policy. In so far as cost estimation is an important element in guiding social policy on smoking it is a matter of concern that no systematic study of social cost exists, with the possible exception of the Leu & Schaub (1984) study for Switzerland.

Having carried out this survey, it is the authors belief that substantial advances could be made in estimating the social costs of smoking by using the conceptual approach outlined in this paper. Such a study could make use of existing empirical data to some extent but would need to collect new data in certain areas. It would, however, do much to remove the extensive errors in the very variable approaches currently available.

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